

Aaron Jacob Sunil

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Professional Summary

Full-stack developer with hands-on experience building LLM-powered applications using Python, FastAPI, and vector databases. Delivered production-ready AI systems across chatbot, RAG, and recommendation domains. Comfortable across the full stack - from PostgreSQL and ChromaDB on the backend to React frontends - with a focus on clean API design and real-world usability.

Academics

APJ Abdul Kalam Technological University (KTU)
B.Tech in Information Technology
CGPA: 7.5 (up to Semester 7); final semester result pending.

Kottayam, India
2022 – 2026

CBSE Board
Higher Secondary Education
Percentage: 78% (12th), 73.2% (10th).

Kuwait
2022

Internship

Full Stack Developer Intern – Sreenarayanaguru Open University
Kollam, Kerala

May 2025 – Jun 2025

Built a bilingual (English/Malayalam) academic chatbot with voice and text interaction, deployed for 500-plus university users. Improved backend response times by implementing query caching and optimizing database access patterns. Integrated JWT-based authentication modules in coordination with the university IT team.

Projects

Lumira – Interactive AI Assistant for Exhibitions
ChromaDB, FastAPI, React

Nov 2025 – Present

Built an AI exhibition assistant using RAG over a ChromaDB vector store, capable of answering 120-plus product queries with contextual accuracy. Tuned chunking strategy and embedding model selection to achieve 92% semantic relevance on a manually evaluated query test set. Reduced end-to-end response time from ~7s to ~5s by switching to async FastAPI endpoints and optimizing vector retrieval; dominant latency is upstream LLM API.

Nutriswap – Intelligent Food Alternative Finder
React, FastAPI, PostgreSQL, Tailwind CSS

Dec 2025 – Present

Built a full-stack health application with a curated database of 100-plus food alternatives, searchable by nutritional profile and allergen flags. Engineered a recommendation engine supporting health-goal filters (Diabetic, Heart-Healthy, Low-Sodium) using PostgreSQL full-text search. Reduced average search latency from ~700ms to ~400ms by adding indexed lookups and optimizing SQLAlchemy query patterns.

InfoBot – AI Chat Assistant for Campus Management
Python, React, FastAPI, MySQL

Feb 2025 – Apr 2025

Built an automated timetable generation module that reduced administrative scheduling time from 2 weeks to under 2 minutes. Handled 100-plus student queries on timetables and grades through a React frontend backed by FastAPI and MySQL. Designed async REST API endpoints in FastAPI to support concurrent user sessions without performance degradation.

TaskIT – Smart Task Management App
Node.js, MongoDB, HTML, CSS, JS

Oct 2024 – Nov 2024

Developed a task management web app with priority-based scheduling using deadline proximity and user-defined urgency weights. Built a modular Node.js backend with MongoDB to support real-time task updates and cross-session state persistence. Designed a single-page UI with dynamic task reordering and status tracking without page reloads.

Technical Expertise

Languages: Python, JavaScript, SQL, HTML, CSS
Frameworks: FastAPI, React, Django, Node.js
Databases: MySQL, MongoDB, ChromaDB, PostgreSQL
Concepts: REST API design, Retrieval-Augmented Generation (RAG), vector search, JWT authentication

Certifications

DeepLearning.AI: ChatGPT Prompt Engineering for Developers.
NPTEL: Fundamentals of Cloud Computing (Elite, 75%).